Visit Summary — Unnathi CNC Technologies Pvt Ltd

Date: 7th February 2026

Visited by: Lakshminarasimhan K & Aravind (Akshaya Createch)

Met: Suresh Kumar S (MD)

Company Profile (from website & visit)

Detail	Information
Company	Unnathi CNC Technologies Pvt Ltd
Address	No. 487, D1 & D2, 13th Cross, 4th Phase, Peenya Industrial Area, Bangalore-560 058
Phone	080-41171792, 9480686747, 9480686743
Email	rfq@unnathicnc.com, sales@unnathicnc.com
Established	2007
Certifications	AS9100D, ISO 9001:2015
Sectors	Aerospace, Medical, Agriculture, Machine Tools
Materials	Carbon alloy, stainless steel, copper, aluminum, brass

Equipment / Machinery

- **9 CNC Machining Centers**
- **19 CNC Turning Centers**
- **3 Grinding Machines**
- Air Compressors, HVAC, Pumps, Coolant systems
- Solar PV installation (on-site)

Key Personnel Identified

Name	Role	Department
Suresh Kumar S	MD / Director	Management
Sandeep G. Parvatikar	Director	Management
Murali	Manager	Management
Rajase Kharkin	Staff	Marketing
Naveen DB	Staff	Design
Nandan Kumar	Staff	Quality
Saadnasukh M	Staff	Turning
Prashant UG PD	Staff	Milling
M. Aral	Staff	Development
Nadhiya S	Staff	Sales

Vilas M.B	Staff	Maintenance
Sumithra	Staff	Polish
Anil Thakur	Staff	Pressing
Jyothi Prasad P	Staff	Grinding
Niranjana C	Staff	Milling
Ravi Kumar	Staff	Wirecut

Visit Outcome

Meeting could NOT be conducted as per the planned agenda.

Primary Reason: There are **NO automatic energy measurement devices/tools** in place at Unnathi CNC. All energy monitoring and auditing is done **entirely manually** (paper-based registers, manual EB bill tracking, handwritten records).

This means:

- Our software system's value proposition of **automating** energy data capture does not directly apply in its current form
- The client needs a solution that works with **manual data entry** workflows first
- There is an opportunity to recommend IoT/smart metering as a future upgrade

---

Documents Collected

We collected the following documents during the visit (ZED Parameter 12 - Energy Management):

12.1 — Energy Policy & Energy Management Plan

Files: `12.1.1.jpeg` , `12.1.2.jpeg` , `12.1.3.jpeg`

Energy Policy (Approved: Sandeep G. Parvatikar, 10/01/2025)

- Policy statement, scope, objectives, commitments defined
- Roles & responsibilities assigned (Top Management, Energy Manager, All Employees)
- Annual review cycle committed

Energy Management Plan (Approved: Sandeep G. Parvatikar, Director, 10/01/2025)

- **Scope:** CNC machining (turning, milling, grinding), ancillary equipment (air compressors, pumps, HVAC), general lighting & facility energy
- **Energy Sources Identified:**
  - Electrical Power (CNC machines, lighting, HVAC, pumps, ancillary)
  - Compressed Air (tool changes, work holding, cleaning)
  - Hydraulics (powering clamps and equipment)
  - Gas/Steam (if applicable — surface treatments, washing stations)
  - Renewables — Solar PV installations
- **Targets:**
  - Short-Term (1-3 months): Minimize idle equipment, switch off when not in use, maintain air compressors, fix air leaks
  - Medium-Term (3-12 months): Reduce energy per part by 5-10%, install LED lighting & sensor controls, optimize machining speeds
  - Long-Term (1-3 years): High-efficiency equipment & servo motors, solar PV expansion, 10-20% overall reduction
- **Action Plan:** 6 action items with owners & timelines (Month 1 to Month 36)
- **Monitoring:** Energy log for equipment & departments, quarterly reviews, benchmarking

---

12.2 — Energy Consumption Data

Files: `12.2.1.jpeg` , `12.2.2.jpeg`

Solar Power Production (Monthly) — Target: 8,000 kWh/month, 5% increase yearly

Month	Solar (kWh)	vs Target
Jul-24	9,136	MORE
Aug-24	7,498	LESS
Sep-24	8,572	MORE
Oct-24	9,110	MORE
Nov-24	8,060	MORE
Dec-24	6,967	LESS
Jan-25	7,456	LESS
Feb-25	8,815	MORE
Mar-25	8,934	MORE
Apr-25	10,866	MORE
May-25	10,301	MORE
Jun-25	9,551	MORE

> ***Remark by Sandeep (30/06/2025):*** *In the months of Aug-24, Dec-25, & Jan-25, solar power production is less than avg due to weather conditions like rain, cyclone effect. Hence corrective action is not required.*

Grid Power Consumption (Monthly) — Target: 22 kWh(?), 10% reduced yearly

Month	Grid (kWh)	vs Target
Jul-24	13,683	LESS
Aug-24	16,360	LESS
Sep-24	16,169	LESS
Oct-24	14,929	LESS
Nov-24	15,999	LESS
Dec-24	17,414	LESS
Jan-25	20,645	LESS
Feb-25	17,100	LESS
Mar-25	17,683	LESS
Apr-25	19,061	LESS
May-25	19,783	LESS
Jun-25	16,186	LESS

> *All months are marked "LESS" than target — meeting 10% reduction goal. Reviewed by Sandeep G. Parvatikar, 30/06/2025.*

Key Energy Observations:

- **Total annual solar generation (Jul-24 to Jun-25):** ~~105,266 kWh~~ (8,772 kWh/month avg)
- **Total annual grid consumption (Jul-24 to Jun-25):** ~~205,012 kWh~~ (17,084 kWh/month avg)
- **Solar contributes ~34% of total energy** — significant renewable share
- **Peak grid month:** Jan-25 (20,645 kWh) — possibly due to lower solar + higher demand
- **Lowest grid month:** Jul-24 (13,683 kWh)

12.3 — Training & Effectiveness Record

File: `12.3.1.jpeg`

Detail	Information
Training Program	Energy Management
Date	08/03/2025
Faculty	Suresh Kumar S
Place	Unnathi CNC Technologies Pvt Ltd
Attendees	14 employees
Topics Covered	Purpose, Scope, Energy Sources, Targets for Energy Efficiency, Action Plan, Monitoring & Verification, Continual Improvement

- All 14 employees rated **"Satisfactory"** in training effectiveness
- Effectiveness measured by **Demonstration/Test (B)** and **On-the-job Assessment (C)** methods
- Assessed by: Sandeep G. Parvatikar (28/03/2025)
- Evidence signed by: Training in-charge, MR, Quality in-charge, Production in-charge (Murali), Director (Sandeep)

12.4 — Energy Management Audit Checklist

Files: `12.4.1.jpeg` , `12.4.2.jpeg` , `12.4.3.jpeg` , `12.4.4.jpeg`

Audited by: Nadhiya, 16/04/2025 **Reviewed by:** Suresh Kumar S, 16/04/2025

Category	Item	Status	Remarks
General Information	Energy policy documented & approved	Yes	
	Energy management team established	Yes	
	Baseline energy consumption recorded	Yes	
	Energy goals and KPIs defined	Yes	
	Compliance with legal/regulatory requirements	Yes	
Energy Data Monitoring	Energy meters installed at major loads	Yes	
	Real-time monitoring system available	Yes	
	Regular energy performance reports prepared	Yes	

	Power factor and load factor tracked	Yes	
Lighting Systems	LED lights installed in all areas	Yes	
	Motion sensors/timers used	**No**	**OI** (Opportunity for Improvement)
	Daylight integration optimized	Yes	
	Lux levels checked and within norms	Yes	
HVAC Systems	Thermostat set at energy-efficient range	**No**	**OI**
	Regular maintenance of HVAC units done	Yes	
	VFDs installed on fans/pumps where feasible	Yes	
	Cooling towers and chillers operating efficiently	N/A	
Motors, Pumps & Compressors	Energy-efficient (IE3/IE4) motors installed	Yes	
	Air leakages in compressor lines checked	Yes	
	Preventive maintenance program in place	Yes	
Process Energy Use	Heat recovery systems used	No	Not applicable
	Insulation of steam/hot water lines intact	No	Not applicable
	Use of renewable energy considered	Yes	Solar panels installed
	Load scheduling during off-peak hours	Yes	
Building Envelope	Roof and wall insulation adequate	Yes	
	Windows and doors sealed properly	Yes	
	Solar heat gain minimized through design	Yes	
Water & Steam Management	Condensate recovery system operational	No	Not applicable
	Water heating optimized for demand	No	Not applicable
	Water leakages identified and fixed	Yes	
Employee Awareness & Training	Energy training provided to staff	Yes	
	Suggestions from employees encouraged	Yes	
	Energy-saving posters/signage displayed	Yes	
Energy Performance Improvement	Energy audits conducted periodically	Yes	
	Energy-saving initiatives implemented	Yes	
	Action plan for continuous improvement available	Yes	

Critical Discrepancies Noted

> ***IMPORTANT:*** *The audit checklist (dated 16/04/2025) states:*
>
> - *"Energy meters installed at major loads?" — **Yes***
> - *"Real-time monitoring system available?" — **Yes***
>
> ***However, during our visit on 07/02/2026, we observed that there are NO automatic energy measurement devices in place. All monitoring is done manually.***
>
> *This is a significant gap — either the meters/monitoring were removed/broken after the audit, or the checklist responses were optimistic. This needs to be clarified with Unnathi.*

Opportunities for Improvement (OI) from Audit

1. **Motion sensors/timers** for lighting — not installed (marked OI)
2. **Thermostat at energy-efficient range** — not set (marked OI)
3. **Automatic energy monitoring** — currently manual; digital system needed for ZED compliance evidence

---

Implications for Our Energy Management System

Given that Unnathi has **no automatic metering**, our system needs to support:

1. **Manual data entry workflows** — operators enter daily meter readings by hand
2. **EB bill data import** — monthly data from electricity bills entered manually
3. **Solar production logging** — manual entry from solar inverter readings
4. **Simple forms** — quick entry screens for daily/weekly/monthly consumption
5. **Calculated analytics** — derive insights from manually entered data (trends, vs-target, deviations)
6. **Print-ready reports** — formatted for ZED assessor review

Future upgrade path:

- Recommend IoT energy meters at key load points
- Integrate with solar inverter data APIs
- Automated data capture to eliminate manual entry

Next Steps

#	Action Item	Owner	Priority	Timeline
1	Clarify discrepancy on energy meters/real-time monitoring claim vs actual manual process	Akshaya Createch	High	Next call
2	Adapt the app to support manual data entry workflows (no IoT dependency)	Akshaya Createch (Dev)	High	1 week
3	Enter actual solar & grid consumption data (Jul-24 to Jun-25) into the system	Akshaya Createch	Medium	After app update
4	Enter employee list (14 staff) into the system	Akshaya Createch	Medium	After app update
5	Import energy policy & management plan content into the system	Akshaya Createch	Medium	After app update
6	Schedule system demo with Suresh & Sandeep (manual-entry focused)	Both	High	Within 2 weeks

7	Recommend IoT/smart meter solutions for future automation	Akshaya Createch	Low	During demo
8	Collect remaining docs: EB bills (actual bills), DG log, factory layout	Unnathi CNC	Medium	Within 2 weeks

---

Document Index

File	ZED Ref	Description
`Unnathi-docs.pdf`	12.x	Complete PDF of all collected documents (10 pages)
`12.1.1.jpeg`	12.1	Energy Policy
`12.1.2.jpeg`	12.1	Energy Management Plan (page 1)
`12.1.3.jpeg`	12.1	Energy Management Plan (page 2 — Approval)
`12.2.1.jpeg`	12.2	Solar Power Production — Monthly Data
`12.2.2.jpeg`	12.2	Grid Power Consumption — Monthly Data
`12.3.1.jpeg`	12.3	Training & Effectiveness Record
`12.4.1.jpeg`	12.4	Energy Audit Checklist — General & Monitoring
`12.4.2.jpeg`	12.4	Energy Audit Checklist — Lighting, HVAC, Motors
`12.4.3.jpeg`	12.4	Energy Audit Checklist — Process, Building, Water
`12.4.4.jpeg`	12.4	Energy Audit Checklist — Training, Performance, Sign-off
`DocScanner 08-Feb-2026 9-00 pm.pdf`	All	ZED Parameters — Full Requirements & Handwritten Notes

---

ZED Certification Parameters — Full Requirements Document

File: `visit/DocScanner 08-Feb-2026 9-00 pm.pdf` (3 pages, scanned 07/02/2026)

This document lists the **complete ZED certification requirements** across all parameters, with handwritten notes captured during the visit. It gives us the full picture of what Unnathi needs beyond just energy management.

Parameters Covered & Status

#	ZED Parameter	Sub-points	Handwritten Notes / Observations
1	**Leadership**	1.1 Roles & Responsibilities defined	
		1.2 Organogram established	
		1.3 Org performance (regulatory, quality, product, safety, environmental) reviewed regularly	
2	**Swachh Workplace**	2.1 SOPs for cleaning & hygiene in place	
		2.2 All equipment, tools, workspaces maintained clean & organized	
		2.3 Cleanliness & hygiene propagated across supply chain	

		2.4 Periodic audit mechanism for cleaning, hygiene & validation	
		2.5 Audit results reviewed periodically	
3	**Occupational Safety**	3.1 Formal safety policy in place	
		3.2 Systems to identify, communicate, train on safety; PPEs ensured	
		3.3 Safety part of KPIs	
		3.4 Safety systems, RCA, CAPA, mock drills, re-training implemented	**"WCA None"** — No Workers' Compensation Act coverage?
		3.5 Periodic review by Senior Management	
5	**Quality Management**	5.1 Quality Policy in place	
		5.2 Quality requirements established & monitored	
		5.3 Training imparted to all relevant people	
		5.4 Periodic quality audits in place	**"Schedule & evidence of conducted to any process"**
		5.5 RCA, CAPA processes in place	
		5.6 QMS review by Senior Management	**"MRM"** (Management Review Meeting)
8	**Planned Maintenance & Calibration**	8.1 PM & calibration plan in place for all machines/equipment/devices	**"Give PM & calibration report"**
		8.2 Maintenance & calibration carried out periodically as per plan	**"Give PM & calibration report"**
		8.3 RCA, CAPA processes in place	**"Same process"**
		8.4 MTTR & MTBF trends reviewed	**"Do this for all machinery & motion — Period 1 year min"**
9	**Process Control**	9.1 All processes proactively planned	
		9.2 SOPs for all critical & non-critical processes established	
		9.3 Monitoring of all critical processes in place	**"Stay inspection & contour report"**
		9.4 RCA, CAPA processes in place with periodic review	**"Same process & MRM"**
10	**Product Quality & Safety**	10.1 All product quality & safety requirements identified	
		10.2 Requirements fulfilled through requisite testing/certification	
		10.3 RCA, CAPA processes in place	
		10.4 Review of product quality & safety by Senior Management	**"MRM"**

12	**Energy Management**	12.1 All energy sources identified & targets set	
		12.2 Energy consumption measured & recorded; corrective actions taken on deviations	
		12.3 Employees trained in energy management	
		12.4 Periodic energy audits in place	
		12.5 RCA, CAPA processes in place	
13	**Environment Management**	13.1 Systems to manage emissions, effluent, waste per regulations	
		13.2 Periodic environmental audits conducted	
		13.3 RCA, CAPA processes in place	
		13.4 People trained on Environment Management Systems	
		13.5 Periodic review by Senior Management	
14	**Measurement & Analysis**	14.1 Processes to track Defects, Rework, Rejection, COPQ, Customer Satisfaction	
		14.2 RCA, CAPA processes in place	**"Same RCA procedure"**
16	**Risk Management**	16.1 Comprehensive Risk Management Plan in place	
		16.2 Risk assessment & mitigation measures for all identified risks	
		16.3 Risk plan reviewed periodically by Senior Management	**"MRM"**
18	**Technology Selection & Upgradation**	18.1 Technology selection planned considering product, market, regulatory, environmental needs	
		18.2 Digitalization, sensors, IoT, smart machines considered for upgradation	**Highlighted — relevant to our system**
20	**Corporate Social Responsibility**	20.1 CSR policy in place	
		20.2 CSR practices defined (governance, labour, environment, fair practices, community)	
		20.3 Periodic review mechanism in place	

Parameters NOT Covered in This Document

Parameters **4, 6, 7, 11, 15, 17, 19** are missing from this document. These likely cover:

- 4: Human Resource Development
- 6: Design
- 7: Supply Chain Management
- 11: Inventory Management
- 15: IPR (Intellectual Property Rights)
- 17: Financial Management
- 19: Marketing / Customer Engagement

Key Takeaways from Handwritten Notes

1. **MRM (Management Review Meetings)** are referenced repeatedly (Parameters 5, 9, 10, 16) — Unnathi uses MRM as their primary mechanism for senior management review across multiple ZED parameters
2. **RCA/CAPA process is shared** — The same Root Cause Analysis and Corrective/Preventive Action procedure is reused across quality, maintenance, process control, energy, and environment parameters ("Same process" / "Same RCA procedure")
3. **PM & Calibration reports needed** (Parameter 8) — They need to provide Preventive Maintenance and calibration reports. Note: "Do this for all machinery & motion — Period 1 year min" suggests this tracking may not be fully in place for all equipment
4. **WCA (Workers' Compensation Act) — "None"** (Parameter 3.4) — This is a potential compliance gap that needs attention
5. **Parameter 18.2 highlighted** — "Digitalization, sensors, IoT, smart machines" — this directly supports our pitch for the energy management system and potential IoT/smart metering recommendations
6. **Parameter 12 (Energy Management)** — Now confirmed as a 5-point requirement including RCA/CAPA (12.5), which our system needs to support

---

Updated Scope Assessment

Based on the full ZED parameters document, our Energy Management system addresses **Parameter 12** directly. However, there are adjacent opportunities:

Our System Feature	ZED Parameter Addressed
Energy source register & targets	12.1
Consumption recording & deviation tracking	12.2
Training module with attendance records	12.3
Internal energy audit module	12.4
RCA/CAPA workflow	12.5 (also reusable for 5.5, 8.3, 9.4, 10.3, 13.3, 14.2)
Dashboard & reports	14.1 (Measurement & Analysis)
Digital system itself	18.2 (Technology & Digitalization)

The **RCA/CAPA module** in our system could potentially serve as the common procedure used across multiple ZED parameters — increasing the system's value proposition significantly.